**P510/1**
**PHYSICS**
**Paper 1**
4 August 2026
**3 hours**

ENTEBBE

**ENTEBBE JOINT EXAMINATION BUREAU**

**Uganda Advanced Certificate of Education**

PHYSICS

Paper 1

**3 hours**

**INSTRUCTIONS TO CANDIDATES:**

*This examination paper consists of* **two** Sections; **A** *and* **B**. *It has* **six** *items*.

Section **A** *has* **two** *compulsory items*.

Section **B** *has* **two** Parts; **I** *and* **II**.

*Respond to* **four** *items in all.*

*Respond to only* **one** *item from each* Part.

*Any additional item(s) responded to, will not be scored.*

*Responses to each item should start on a fresh page.*

*Silent non-programmable scientific calculators may be used.*

## SECTION A

*Respond to* **both** *items in this* Section.

**ITEM 1**

A recent road safety audit by the Ministry of Works and Transport (MoWT) has identified a high frequency accident 'black spot' at a sharp corner within a busy trading centre. Oncoming drivers from opposite directions cannot see each other at all, leading to frequent head-on collisions. The road safety committee has acquired two different types of curved mirrors each of radius of curvature 2.0 m and a plane mirror all of the same surface area. The committee wants to install a suitable mirror to view oncoming hazards before entering the sharp corner. Each of the mirrors is tested to check its field of view and the committee wants to understand which of the mirrors has a wide field of view before installation is made.

Further tests are carried out for each mirror on a test vehicle, which is 1.5 m tall, drives along the highway and stops at a distance of 3.0 m directly in front of the prototype-installed safety-mirror. The committee needs to evaluate the performance of each of the prototype mirrors to ensure that it provides a clear image.

The audit has noticed that during heavy rains, water droplets blur the mirror glass and darkness reduces visibility; making it impossible for the oncoming drivers to see the black spot in the installed mirror. The audit wants to understand how the selected mirror type can be designed to ensure high continuous reliability and high visibility for night road users.

**Tasks:**

(a) Respond to the concerns put across by the committee

(b) Propose a suitable mirror design to address the audit's concern.

**ITEM 2**

A certain village located in a mountainous region is experiencing landslides and floods, posing severe threats of property loss and crop destruction. The village council wishes to install a communication alerting system that can help transmit timely warnings to farmers and the entire community.

Two potential options are being evaluated. System **A** uses a siren that generates waves at 212.5 Hz and is considered effective if its wavelength exceeds 2.0 m. System **B** uses a battery-powered FM transmitter to send signals to household radios operating at 3.53 m band, remaining effective if operated below 85 MHz. Before making a choice, the council wants to understand the exact nature of waves emitted by these two systems and how they propagate through the community. However, due to recent heavy storms,

the signals generated by the installed system suffered severe atmospheric interference and terrain blockage, failing to reach households in deeper valleys. A regional telecommunication engineer stepped in and proposed a modern alternative system **C**. This system bypasses atmospheric issues by transmitting high-speed digital light signals through a buried optical network cable directly to a central alert terminal in the village centre. The community is now debating whether this new upgrade will truly overcome the previous failure and ensures no significant loss of signal.

Hint**:** **S**peed of light in air = $340 ms^{-1}$
Speed of light = $3.0 \times 10^{8} ms^{-1}$

**Task:**

(a) Help the council to analyse the right selection between systems **A** and **B**.

(b) Help the council to understand how the proposed modern alternative system ensures no significant loss of signal across the network.

## SECTION B

## PART I

*Respond to* **one** *item from this* Part

**ITEM 3**

You are a Physics intern at the National Forensic Laboratory helping banks to identify forged banknotes. Genuine notes feature a transparent, micro-etched security laminate that acts as a Young's double-slit system. When a calibration laser of wavelength $5.89 \times 10^{-7}$ m passes through this laminate onto a laboratory screen fixed 1.60 m away, it creates an interference pattern. Your laboratory measuring rig currently measures the distance from the central maximum to the fourth bright fringe as exactly 16.0 mm.

The bank manager has raised concerns regarding the scientific validity and legality of this set-up. He points out that court cases require proof of how these invisible micro-slits produce visible patterns. Furthermore, he presents the official legal standards, which dictate that a testing rig is only certified for legal evidence if its calculated slit separation falls strictly within the range $2.30 \times 10^{-4}$ to $2.4 \times 10^{-4}$ m. He is worried your rig might fail this legal audit.

Finally, the manager notices that while this bulky 1.60 m laboratory set-up works for experts, local market vendors need a compact, handheld scanner to protect their businesses. However, he is concerned that if you drastically reduce the screen distance, **D** to make the device small, the interference fringes will become too marrow to see. He needs to know how you can modify the device parameters to keep the fringes wide enough for the vendors to easily see and verify.

**Task:**

(a) Respond to the concern put across by the bank manager.

(b) Propose the modifications to the design for a functional hand-held scanner for the local market vendors.

**ITEM 4**

A local electronic plant is responsible for assembling smartphone screens for the Ugandan market. The Quality Assurance team uses the optical interference method to verify a new supplier's component standard before mass production begins. The optical spacer foil being used has a uniform thickness of exactly $1.7 \times 10^{-5}$ m to maintain optimal screen sensitivity. Any significant deviation from this thickness will cause the screen to malfunction, leading to major financial losses and electronic waste.

During quality testing, two flat glass plates are placed in contact at one end and separated by a piece of the supplier's thin metal foil at the other end, trapping air film between them. The foil is placed at a distance 75 mm from the line of contact. When monochromatic light of wavelength $5.6 \times 10^{-7}$ m is shone directly down onto the plates, a parallel interference fringes with a consecutive separation distance of 1.2 mm are observed. The team wants to understand how these fringes are formed. The team is also concerned about whether this batch of foil meets their manufacturing standards and if not what can be done to ensure they avoid the major financial losses and electronic waste.

Furthermore, the team wants to eliminate manual inspection delays by integrating a rapid, non-destructive optical testing procedure into their production line that uses thin film principles to verify the optical flatness and smoothness of incoming glass slides before assembly.

**Task:**

(a) Address the concerns of the quality team.

(b) Help the quality team with the design of the optical testing required for the production line.

## PART II

*Respond to* **one** *item from this* Part

**ITEM 5**

A group of Physics students is conducting a project on wave properties of light. While walking to school on a rainy day, they notice a multi-coloured pattern dancing on a thin layer of engine oil floating on a puddle of water in the road. They decide to investigate whether a thin air-film exhibits similar effects under controlled laboratory conditions. They set up an air-film between two clean glass plates in contact at one end with thin

foil at the other. When they direct light of wavelength 6.0 x $10^{-7}$ m normally onto one of the glass plates and look through a travelling microscope, they are surprised to see distinct bright and dark fringes with the first dark fringe at the point of contact of the glass plate. They are surprised to see a different situation of patterns and want to understand these observations.

As they move the microscope down the air-film, they note the position of the twentieth dark fringe and thirtieth bright fringe. One learner claims that the thickness of air-film at the two positions is obtained using exactly the same method. This brings mixed reaction from the other learners. The only way to address this is to determine thickness of the air-film at both positions so that they are able to evaluate the validity of the claim.

Soon thereafter, their teacher gives them an assignment to extend the project by replacing the air-film with water of refractive index 1.33. The teacher wants them to account for the predictions of the observed effect, check whether the thickness of the new water film at the twentieth dark fringe would remain the same and then advise the manufacturers of glass on how to design an anti-reflective coating for microscopic glass slides to test for optical flatness.

**Task:**
Help the students to
(a) address their concerns during the project work.

(b) respond to the assignment the teacher gave them.

**ITEM 6**
Your class was taken to one of the battery companies in town. Of late, customers are complaining about the quality of the batteries produced. This prompted the company manager to design an investigation to verify customers' complaints using two quality checks.

In the first check, a battery is connected in series with two resistors of resistances 10 Ω and 25 Ω. The potential difference across the 10 Ω resistor was balanced against the potential drop across 715 mm of a uniform potential wire carrying a steady current. The 25 Ω resistor was then replaced with 33 Ω resistor while keeping the 10 Ω resistor unchanged. The length of the same potentiometer wire carrying the same amount of current required to balance the potential difference across the 10 Ω resistor was found to be 582 mm.

In the second check, an ammeter is connected to a 50 Ω resistor and a setup which has a soft iron ring with two coils A and B. The battery is to be connected to an inverter which changes d.c voltage to a.c voltage of r.m.s value equal to the e.m.f of the battery

Coil B has turns four times that of coil A. Coil A is to be connected across the inverter while B is to be connected in series with the ammeter and the resistor. The ammeter reading of 0.96 A is expected. The manager told the learners that he anticipates that the resistor will heat up after some time.

The manager further guided that for their type of battery to be pronounced to be good, it should have an internal resistance not exceeding 1.0 Ω and its e.m.f should be in the range of 11.0 – 13.0 V. The manager shared this information with the learners as a way of challenging them to see if they mastered the concepts they had already learnt in classroom.

**Task:**

(a) Analyse the information provided to verify customers' complaints.

(b) Interpret the operation and a real-life application of the device (having A and B).



**END**